IGNITE INNOVATION CHALLENGE

Problem Statement Briefing

TRUSTED AI • INTELLIGENT OPERATIONS • DIGITAL TRUST • ADAPTIVE BANKING

Build practical solutions for the changing financial-services landscape.



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STUDENT BRIEFING

Choose a challenge that matches your team's strengths

01 **Trusted AI Agents**

Applied AI • Data • Governance

02 **Intelligent Operations**

Software • Automation • Analytics

03 **Digital Trust**

Cybersecurity • AI • Digital Forensics

04 **Adaptive Banking**

UX • Accessibility • Software

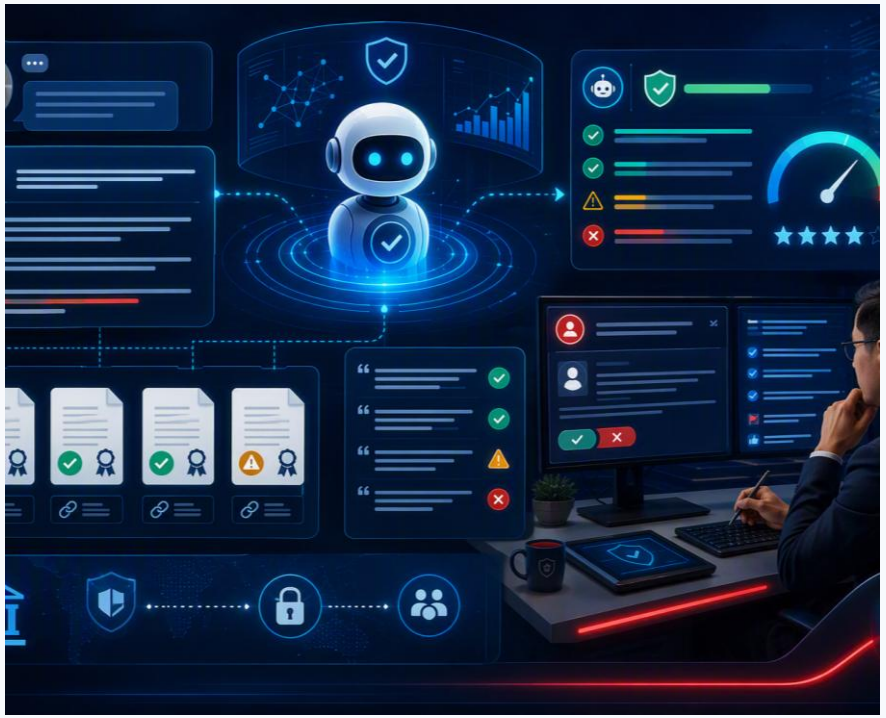
Each brief follows the same headers: Problem Overview, Objectives, Scope, Technology Stack and Possible Expected Outcomes.

PROBLEM STATEMENT 1

Trusted AI Agents: Can We Believe the Answer?

1. Problem Overview

Banks are moving from AI tools that generate text to AI agents that can retrieve information, use tools and support work. An agent may still provide information that is inaccurate, incomplete, outdated or unsupported.



2. Objectives

Develop a solution that helps users assess and improve AI-generated answers for reliability, relevance, groundedness and alignment to user intent.

PROBLEM STATEMENT **How might OCBC define, evaluate and improve an AI agent's answers so employees can judge whether they are trustworthy?**

PROBLEM STATEMENT 1

Trusted AI Agents: Can We Believe the Answer?

3. Scope

Possible Features (students may propose other features)

Define measures of agent “correctness”
Compare answers with approved sources
Verify citations and knowledge freshness
Detect unsupported claims or inconsistencies
Display confidence or reliability indicators
Route uncertain outputs for human review

Out of Scope

Production banking deployment
Use of customer or confidential bank data
Training a foundation model from scratch

4. Technology Stack

Students may explore any technology

Front-End: React, Vue or Streamlit
Back-End: Python, Node.js or .NET
AI: Azure OpenAI, Semantic Kernel or LangChain
Data: Approved sample or synthetic knowledge base
Analytics: Power BI, Grafana or custom dashboard

5. Possible Expected Outcomes

A demonstrable, user-centred prototype

Evaluate an AI answer – User
View supporting sources and explanations – User
Review flagged answers – Human Reviewer
Track reliability trends – Administrator
Recommend agent improvements – System

PROBLEM STATEMENT 2

Intelligent Operations: From Automation to Autonomous Resolution

1. Problem Overview

Operations and technology teams manage repeated checks, exceptions, handovers, reconciliations, approvals and follow-ups. Automating a single step does not necessarily resolve the underlying case.



2. Objectives

Develop an intelligent workflow that detects, investigates and resolves suitable operational exceptions faster while preserving accuracy, control and a clear audit trail.

PROBLEM STATEMENT **How might OCBC use AI agents, automation and data to resolve routine operational exceptions, not merely describe them?**

PROBLEM STATEMENT 2

Intelligent Operations: From Automation to Autonomous Resolution

3. Scope

Possible Features (students may propose other features)

Detect and classify exceptions
Collect evidence from sample sources
Identify missing information
Recommend causes and next actions
Resolve low-risk cases or escalate to a human

Out of Scope

Connection to live banking systems
Autonomous high-risk decisions
Use of real customer or transaction data

4. Technology Stack

Students may explore any technology

Front-End: React, Angular or Streamlit
Back-End: Python, Java, Node.js or .NET
Automation: Workflow engine or RPA simulator
AI: Agent framework and rules engine
Data: Synthetic cases and event logs

5. Possible Expected Outcomes

A demonstrable, user-centred prototype

Detect an exception – System
Investigate and assemble evidence – Agent
Approve, override or escalate – Reviewer
Track actions and decisions – Administrator
Measure resolution time and manual steps – Process Owner

PROBLEM STATEMENT 3

Digital Trust: Protecting Customers from AI-Powered Scams

1. Problem Overview

Generative AI can produce convincing messages, voices, images and videos. This can make impersonation and social-engineering attempts harder for customers and employees to recognise.



2. Objectives

Develop a solution that helps people recognise, verify and respond to suspected AI-enabled impersonation or scam attempts before money or sensitive information is lost.

PROBLEM STATEMENT **How might OCBC create a useful moment of friction when acting too quickly could cause customer harm?**

PROBLEM STATEMENT 3

Digital Trust: Protecting Customers from AI-Powered Scams

3. Scope

Possible Features (students may propose other features)

Analyse simulated messages or call context
Identify warning signals and inconsistencies
Provide an official verification journey
Introduce a pause-and-verify step
Explain risks and recommend safe actions

Out of Scope

Surveillance of real customers
Collection of real biometric information
Blocking legitimate transactions without review

4. Technology Stack

Students may explore any technology

Front-End: Mobile or responsive web prototype
Back-End: Python, Node.js or .NET
AI: Classification, anomaly detection or multimodal analysis
Security: Identity and verification simulation
Data: Synthetic scam and legitimate scenarios

5. Possible Expected Outcomes

A demonstrable, user-centred prototype

Check suspicious communication – Customer
Understand warning signals – Customer
Verify through an official channel – Customer
Triage a reported case – Employee
Record evidence and escalation – System

PROBLEM STATEMENT 4

Adaptive Banking: One Experience for Every Customer

1. Problem Overview

A single static digital interface cannot serve every customer equally well. Customers may differ in language, accessibility needs, digital confidence and familiarity with a transaction.



2. Objectives

Develop a digital banking experience that adapts responsibly to customer-selected needs and circumstances while remaining secure, understandable and easy to control.

PROBLEM STATEMENT **How might a banking experience adapt to the customer without making assumptions about the customer?**

PROBLEM STATEMENT 4

Adaptive Banking: One Experience for Every Customer

3. Scope

Possible Features (students may propose other features)

Adjust language and reading complexity
Offer voice, text and visual interaction
Guide unfamiliar transactions step by step
Provide accessible error recovery
Enable safe hand-off to human assistance

Out of Scope

Inferring sensitive customer characteristics
Using personalisation to reduce access to services
Connection to live customer accounts

4. Technology Stack

Students may explore any technology

Front-End: React, Flutter or accessible web technologies
Back-End: Node.js, Python or .NET
AI: Language simplification, voice or translation
Design: WCAG-informed interaction patterns
Data: Synthetic personas and journeys

5. Possible Expected Outcomes

A demonstrable, user-centred prototype

Select assistance preferences – Customer
Complete a guided journey – Customer
Switch interaction mode – Customer
Request human assistance – Customer
Review usability and accessibility insights – Designer

STUDENT BRIEFING

Submission principles

SOLVE A REAL PROBLEM

Demonstrate why the selected problem matters and who benefits.

BUILD, DO NOT JUST DESCRIBE

Show a working journey, prototype or testable proof of concept.

DESIGN FOR TRUST

Include clear controls, explanations, review or escalation where needed.

USE SAFE DATA

Use approved sample or synthetic data, not customer or confidential information.

TEST FAILURE SCENARIOS

Show what happens when data is missing, confidence is low or the system is wrong.

MEASURE VALUE

Define a meaningful outcome such as quality, speed, inclusion, safety or reduced effort.

Your goal is not to build the biggest solution. Build the clearest, safest and most convincing one.

STUDENT BRIEFING

Judging Criteria

- **Problem Understanding and Relevance**
- **Innovation and Creativity**
- **Technical Execution**
- **User Experience and Design**
- **Quality of Presentation**
- **Collaboration & Communication**